

1 SQL & Databases – Complete Beginner Notes

▼ 1 What is Data?

Data is everywhere.

Anything that describes something is data.

Examples of Data:

- Your **first name**
- Your **mobile number**
- Contacts, photos, messages inside your phone
- Car sensors generating speed, fuel, and location data
- Bank transactions and financial statements
- Sales, customers, products in companies

👉 **Individuals** usually store data in:

- Excel sheets
- Text files
- PDFs

👉 **Companies** generate **huge volumes of data:**

- Customer data
- Sales data
- Product data
- Employee data
- Logs, analytics, transactions

So the big question is:

How do companies store and manage such massive data efficiently?

▼ 2 What is a Database?

A **database** is a **container used to store data in an organized way**.

Unlike simple files and folders, a database:

- Organizes data into structured formats
- Makes data easy to **store, search, update, and manage**
- Handles **millions or even billions of records**

Simple Definition:

| A database is an organized container for storing data.

▼ 3 Why Not Just Use Files?

You *can* use files personally, but companies **cannot rely on files**.

Problem with Files:

Imagine someone asks:

“Find my total spending for last year.”

Using files:

- Open many files
- Search manually
- Combine values
- Very slow and messy

Using a database:

- Ask **one question**
- Database instantly returns the result

Other Reasons Databases Are Better:

- Handle **huge amounts of data**
 - Faster searching and reporting
 - **More secure**
 - Control **who can access what**
 - More professional and reliable
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▼ 4 What is SQL?

SQL stands for **Structured Query Language**.

What SQL Does:

- SQL is the **language used to talk to databases**
- You ask questions → Database gives answers

Example:

```
SELECT * FROM Customers;
```

📌 Some people say "**SQL**", others say "**Sequel**".

Both are correct.

▼ 5 How Databases Are Used in Companies

In real companies:

- Employees write SQL queries
- Applications (web/mobile apps) send SQL queries
- Reporting tools (Power BI, Tableau) generate SQL
- Thousands or millions of SQL requests happen daily

So we need a **manager** to handle all this.



6 What is DBMS (Database Management System)?

DBMS is **software that manages the database.**

Responsibilities of DBMS:

- Handles multiple SQL requests
- Decides execution order (priority)
- Manages security and permissions
- Ensures data consistency

 Examples of DBMS:

- Microsoft SQL Server
 - MySQL
 - PostgreSQL
 - Oracle
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▼ 7 What is a Server?

A **server** is a **powerful computer that runs 24/7.**

Why not use a normal PC?

- PCs are weak
- PCs go offline
- Databases must always be available

Servers can be:

- **On-premise** (inside company)
 - **Cloud servers** (AWS, Azure, GCP)
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▼ 8 Summary So Far

Component	Meaning
Database	Stores data
SQL	Language to talk to data
DBMS	Software that manages database
Server	Powerful machine where DB runs

▼ 9 Types of Databases

▼ 1. Relational Database (SQL Database)

- Data stored in **tables**
- Rows and columns
- Tables related to each other

 Examples:

- Microsoft SQL Server
 - MySQL
 - PostgreSQL
 - Oracle
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▼ 2. Key-Value Database (NoSQL)

- Data stored as **key** → **value**
- Like a dictionary

 Examples:

- Redis
 - Amazon DynamoDB
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▼ 3. Column-Based Database (NoSQL)

- Data grouped by columns
- Optimized for analytics and big data

 Examples:

- Cassandra
 - Amazon Redshift
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▼ 4. Graph Database (NoSQL)

- Focuses on **relationships**
- Nodes and edges

 Example:

- Neo4j
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▼ 5. Document Database (NoSQL)

- Data stored as documents (JSON)
- Flexible structure

 Example:

- MongoDB
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▼ SQL vs NoSQL

- **Relational databases** → **SQL**
- **Key-Value, Column, Graph, Document** → **NoSQL**

 This course focuses on **SQL Relational Databases**, especially **Microsoft SQL Server**.

▼ 10 Database Hierarchy (Structure)

1. Server



2. Databases



3. Schemas



4. Tables



5. Rows & Columns

▼ 11 What is a Schema?

A **schema** is a **logical container** to organize database objects.

Example:

- **Sales** schema → orders, invoices
- **HR** schema → employees, payroll

Helps manage large databases neatly.

▼ 1 2 What is a Table?

A **table** is like an Excel sheet.

Table Components:

- **Columns (Fields)** → Define type of data
- **Rows (Records)** → Actual data

Example:

CustomerID	Name	Age	City
1	Maria	30	London
2	John	25	New York

▼ 1 3 Primary Key

A **Primary Key**:

- Uniquely identifies each row
- No duplicates
- No NULL values

Example:

- CustomerID

📌 Think of it as a **fingerprint** for a record.

▼ 1 4 Data Types

Each column stores a specific **data type**.

Common Data Types:

- `INT` → Numbers
- `DECIMAL` → Numbers with decimals
- `CHAR(n)` → Fixed-length text
- `VARCHAR(n)` → Variable-length text
- `DATE` → Date
- `TIME` → Time
- `DATETIME` → Date + Time

CHAR vs VARCHAR:

- `CHAR(5)` → Always uses 5 characters
 - `VARCHAR(5)` → Uses only required space
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▼ 1 5 SQL Commands (Categories)

▼ 1. DDL – Data Definition Language

Used to define structure.

Command	Purpose
CREATE	Create table/database
ALTER	Modify structure
DROP	Delete object

▼ 2. DML – Data Manipulation Language

Used to manage data.

Command	Purpose
INSERT	Add new data
UPDATE	Modify data
DELETE	Remove data

▼ 3. DQL – Data Query Language

Used to query data.

Command	Purpose
SELECT	Retrieve data

📌 We spend most time learning **SELECT**.

▼ 4. TCL (Transaction Control Language) ★ Important

Controls transactions.

Commands

```
COMMIT ROLLBACK SAVEPOINT
```

▼ 5. DCL (Data Control Language) ★ Important

Controls permissions and security.

Commands

```
GRANT REVOKE
```

Correction: Your original notes only covered DDL, DML, and DQL. In practice, SQL is commonly divided into **five** command categories: DDL, DML, DQL, DCL, and TCL.

▼ 1 6 Why Learn SQL?

▼ 1 To Talk to Data

- Most company data is in databases
- SQL is the only way to access it

▼ 2 High Job Demand

SQL is required for:

- Software Developers
- Data Analysts
- Data Engineers
- Data Scientists

▼ 3 Industry Standard

SQL is used in:

- Power BI
 - Tableau
 - Spark
 - Kafka
 - Synapse
 - Many modern tools
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▼ Final Summary

- Data is everywhere
 - Databases store data efficiently
 - SQL is the language to interact with databases
 - DBMS manages everything
 - Servers host databases
 - SQL databases are structured and powerful
 - SQL is essential for data and software careers
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